

NWIS

DAILY DRILLING REPORT

Doc ID
W004-DDR-18

Well: Alpha-04 | Well ID: W004 | Field: Field_Alpha | Date: 22-11-2024

PRIMARY ISSUE CATEGORY: CEMENTING FAILURE

1. REPORT SUMMARY

Report No.	18	Date	22-11-2024
Start Depth	2600 m	End Depth	2600 m
Progress (24 hr)	0 m	Formation	Formation_X (upper, at shoe)
Rig	Rig Sagar-4	Mud Type	WBM (displaced ahead of cement)

2. MUD PROPERTIES

MW (SG)	PV (cP)	YP (lb/100ft2)	Gel 10s/10m	pH	Solids (%)
1.20	26	16	7/18	9.5	5.8

3. CEMENT JOB PUMP LOG

Stage	Fluid	Volume (bbl)	Density (ppg)	Rate (bpm)	Pressure (psi)
1	Spacer	30	9.2	6.0	850
2	Lead slurry (Class G)	120	13.8	5.5	1050
3	Tail slurry (Class G)	80	15.8	4.0	1400
4	Tail slurry (losses began)	55 lost	15.8	2.5	980 (declining)
5	Displacement	95	8.4 (mud)	5.0	1250

4. CEMENT BOND EVALUATION (POST-JOB CBL/VDL)

Interval (m)	Bond Quality	Amplitude Attenuation	Interpretation
2500-2540	Good	>70%	Adequate isolation
2540-2600	Poor	<30%	Channeling suspected; remedial squeeze required
2565-2580	Very Poor	<15%	Confirmed annular gas channel

5. OPERATIONS TIMELINE (24 HR)

Time	Duration (hr)	Activity
00:00-02:00	2.0	Rig up cementing unit; pressure test lines to 5000 psi
02:00-03:30	1.5	Pump spacer and lead slurry (150 bbl combined)

03:30-04:15	0.75	Pump tail slurry; full returns maintained
04:15-04:55	0.67	Losses detected during tail slurry; rate reduced, job continued
04:55-06:00	1.08	Displace to bump plug; pressure test casing
06:00-16:00	10.0	WOC (wait on cement); rig down cementing unit
16:00-18:00	2.0	Run CBL/VDL log; poor bond identified 2540-2600 m
18:00-24:00	6.0	Mobilize squeeze cementing equipment; prep perforate-and-squeeze plan

6. EVENT / INCIDENT DETAIL

At 2600 m (shoe interval, losses at 2570-2600 m), cement returns were lost approximately 40 minutes into pumping the tail slurry, with an estimated 55 bbl lost to the loss-prone sand streak above the shoe, later confirmed by CBL/VDL as a poor bond with an annular gas channel at 2565-2580 m.

Action Taken:

1. Pump rate was reduced from 4.0 to 2.5 bpm once losses were observed.
2. Job was continued to completion with a top-up job to fill the annulus as much as possible.
3. CBL/VDL log was run after WOC to evaluate bond quality across the shoe interval.
4. Remedial squeeze cementing was performed at 2570 m: 25 bbl Class G neat slurry (15.8 ppg) via perforate-and-squeeze.

Result: Follow-up CBL/VDL showed improved bond (>65% amplitude attenuation) across the treated interval; casing pressure test to 3500 psi held successfully.

NPT Incurred: 34.0 (10.0 original job losses + 24.0 remedial squeeze) hours

Engineer Remark: This is the first confirmed cementing failure in Field_Alpha tied directly to the known Formation_X loss zone; recommend mandatory LCM-additive lead slurry for any future shoe set in this interval.

7. PERSONNEL ON SITE

Company Man	R. Verma	Drilling Supervisor	D. Bhatt
Mud Engineer	S. Iyer	Toolpusher	N. Joshi

This Daily Drilling Report is a synthetically generated historical record produced for the NWIS prototype (W004). Not associated with an actual drilling operation.